ARCHITECTURAL TECHNOLOGY

Complete Lecture Notes and Study Guide

ND/HND Level • Building Design, Construction and Technology

Prepared as a comprehensive academic study resource

1. Introduction to Architectural Technology

Core Concepts

Architectural technology is the technical study of how buildings are designed, detailed, constructed, serviced and maintained. It connects architectural ideas with materials, construction methods, structural systems, environmental performance and building regulations. The architectural technologist must understand both drawings and physical construction. A good building should be safe, functional, durable, comfortable, economical and appropriate to its environment.

Technical Application and Study Notes

Architectural technology covers technical drawing, building materials, construction processes, structural principles, building services, environmental design, digital tools and project coordination. It also requires communication with architects, engineers, quantity surveyors, builders, contractors and clients. The discipline is important because a design that cannot be safely or economically constructed is incomplete.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

1. Introduction to Architectural Technology — Extended Notes

Detailed Discussion

Architectural technology is the technical study of how buildings are designed, detailed, constructed, serviced and maintained. It connects architectural ideas with materials, construction methods, structural systems, environmental performance and building regulations. The architectural technologist must understand both drawings and physical construction. A good building should be safe, functional, durable, comfortable, economical and appropriate to its environment. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

2. History and Development of Architecture

Core Concepts

Early buildings were produced from locally available materials such as earth, stone, timber, reeds and thatch. Ancient civilizations developed sophisticated construction methods. Egyptian architecture is associated with monumental stone structures; Greek architecture developed classical orders and proportion; Roman builders advanced the use of arches, vaults, domes and concrete.

Technical Application and Study Notes

Later periods introduced Romanesque, Gothic, Renaissance, Baroque and other architectural traditions. The Industrial Revolution transformed construction through mass production, iron, steel, glass and reinforced concrete. Modern and contemporary architecture increasingly combine digital design, advanced engineering, prefabrication and sustainability.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

2. History and Development of Architecture — Extended Notes

Detailed Discussion

Early buildings were produced from locally available materials such as earth, stone, timber, reeds and thatch. Ancient civilizations developed sophisticated construction methods. Egyptian architecture is associated with monumental stone structures; Greek architecture developed classical orders and proportion; Roman builders advanced the use of arches, vaults, domes and concrete. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

3. Elements and Principles of Design

Core Concepts

The elements of architectural design include point, line, plane, form, space, colour, texture, pattern and scale. These elements are combined to communicate function and visual character. Space is particularly important because architecture organizes activities within three-dimensional environments.

Technical Application and Study Notes

Major design principles include balance, rhythm, emphasis, proportion, scale, hierarchy, unity, contrast and harmony. Functional planning should consider users, furniture, circulation, privacy, lighting, ventilation, accessibility and relationships between spaces.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

3. Elements and Principles of Design — Extended Notes

Detailed Discussion

The elements of architectural design include point, line, plane, form, space, colour, texture, pattern and scale. These elements are combined to communicate function and visual character. Space is particularly important because architecture organizes activities within three-dimensional environments. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

4. Architectural Drawing and Graphics

Core Concepts

Architectural drawings communicate design and construction information. Common drawings include site plans, floor plans, elevations, sections, roof plans, details, schedules and presentation drawings. A floor plan shows the arrangement of spaces horizontally, an elevation shows a vertical face, and a section reveals internal construction relationships.

Technical Application and Study Notes

Technical drawings use title blocks, scales, dimensions, symbols, notes and standardized line conventions. Freehand sketching develops observation and design communication. Digital drafting improves precision, editing and documentation, but technical knowledge remains essential.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

4. Architectural Drawing and Graphics — Extended Notes

Detailed Discussion

Architectural drawings communicate design and construction information. Common drawings include site plans, floor plans, elevations, sections, roof plans, details, schedules and presentation drawings. A floor plan shows the arrangement of spaces horizontally, an elevation shows a vertical face, and a section reveals internal construction relationships. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

5. Building Materials

Core Concepts

Building materials are selected according to strength, durability, availability, cost, appearance, maintenance, fire performance and environmental impact. Common materials include concrete, steel, timber, masonry, glass, aluminium, tiles, plaster, paint and roofing products.

Technical Application and Study Notes

Concrete is widely used in foundations, floors and structural frames. Steel provides high strength and is used in reinforcement and structural frames. Timber can be used in roofs, doors, finishes and structural systems. Masonry provides wall construction, while glass provides daylight and views. Material properties such as strength, density, thermal conductivity, water absorption and durability influence performance.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

5. Building Materials — Extended Notes

Detailed Discussion

Building materials are selected according to strength, durability, availability, cost, appearance, maintenance, fire performance and environmental impact. Common materials include concrete, steel, timber, masonry, glass, aluminium, tiles, plaster, paint and roofing products. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

6. Site Investigation and Preparation

Core Concepts

Site investigation is carried out before construction to understand the physical and environmental conditions of a site. It may include measurement, topography, soil investigation, drainage, vegetation, access, existing buildings and utility services.

Technical Application and Study Notes

Site preparation may involve clearing, setting out, temporary drainage, access and removal of unsuitable material. Setting out transfers dimensions from drawings to the ground. Benchmarks, profiles and accurate measurements help ensure that the building is positioned correctly.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

6. Site Investigation and Preparation — Extended Notes

Detailed Discussion

Site investigation is carried out before construction to understand the physical and environmental conditions of a site. It may include measurement, topography, soil investigation, drainage, vegetation, access, existing buildings and utility services. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

7. Foundations

Core Concepts

Foundations transfer building loads safely to the ground and help control settlement. Selection depends on soil conditions, loads, groundwater, site constraints and economics. Shallow foundations include strip, pad and raft foundations.

Technical Application and Study Notes

Deep foundations such as piles may be used where near-surface soils are unsuitable or where structural requirements demand greater depth. Foundation construction requires safe excavation, suitable concrete and reinforcement, moisture protection, correct levels and proper backfilling.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

7. Foundations — Extended Notes

Detailed Discussion

Foundations transfer building loads safely to the ground and help control settlement. Selection depends on soil conditions, loads, groundwater, site constraints and economics. Shallow foundations include strip, pad and raft foundations. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

8. Walls and Wall Construction

Core Concepts

Walls may be load-bearing or non-load-bearing. They provide enclosure, privacy, weather protection, fire separation, acoustic control and thermal performance. Masonry walls commonly use blocks or bricks joined with mortar.

Technical Application and Study Notes

Good wall construction requires correct bonding, alignment, verticality, jointing and support over openings. Finishes include plastering, rendering, painting, tiling, cladding and exposed masonry. Damp-proofing and drainage details help control moisture.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

8. Walls and Wall Construction — Extended Notes

Detailed Discussion

Walls may be load-bearing or non-load-bearing. They provide enclosure, privacy, weather protection, fire separation, acoustic control and thermal performance. Masonry walls commonly use blocks or bricks joined with mortar. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

9. Floors and Floor Finishes

Core Concepts

Floors provide level surfaces, carry imposed loads and separate building levels. Ground floors may use compacted fill, a concrete slab, damp-proof membrane and a selected finish. Suspended floors may use timber, concrete, steel or proprietary systems.

Technical Application and Study Notes

Floor finishes include ceramic tiles, terrazzo, concrete, timber, vinyl and carpet. Selection depends on traffic, moisture, appearance, maintenance, comfort and cost. Proper levels, joints and moisture control are important for durable floors.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

9. Floors and Floor Finishes — Extended Notes

Detailed Discussion

Floors provide level surfaces, carry imposed loads and separate building levels. Ground floors may use compacted fill, a concrete slab, damp-proof membrane and a selected finish. Suspended floors may use timber, concrete, steel or proprietary systems. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

10. Roofs and Roofing Systems

Core Concepts

Roofs protect buildings from rain, solar heat and other environmental conditions. Common forms include flat, mono-pitch, gable, hip, mansard and curved roofs. Roof form influences drainage, structural behaviour, usable space and appearance.

Technical Application and Study Notes

Roof components can include rafters, trusses, purlins, decking, underlay, battens, roof covering, gutters, valleys, ridges and flashings. Roofing materials include metal sheets, tiles, fibre-cement products and membrane systems. Junctions and penetrations require careful detailing.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

10. Roofs and Roofing Systems — Extended Notes

Detailed Discussion

Roofs protect buildings from rain, solar heat and other environmental conditions. Common forms include flat, mono-pitch, gable, hip, mansard and curved roofs. Roof form influences drainage, structural behaviour, usable space and appearance. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

11. Doors and Windows

Core Concepts

Doors provide access, privacy, security, fire separation and controlled circulation. Common types include hinged, sliding and folding doors. Windows provide daylight, ventilation, views and sometimes emergency escape.

Technical Application and Study Notes

Door and window selection should consider weather resistance, thermal performance, acoustic control, security, accessibility, glazing and maintenance. Correct installation and sealing are essential for long-term performance.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

11. Doors and Windows — Extended Notes

Detailed Discussion

Doors provide access, privacy, security, fire separation and controlled circulation. Common types include hinged, sliding and folding doors. Windows provide daylight, ventilation, views and sometimes emergency escape. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

12. Stairs and Circulation

Core Concepts

Stairs provide vertical circulation. Main components include treads, risers, landings, handrails, balustrades and supporting structures. Consistent dimensions improve safety and comfort.

Technical Application and Study Notes

Other circulation systems include ramps, lifts and escalators. Corridors should be wide enough for expected movement and should provide clear routes. Accessibility requires attention to gradients, handrails, door widths and turning spaces.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

12. Stairs and Circulation — Extended Notes

Detailed Discussion

Stairs provide vertical circulation. Main components include treads, risers, landings, handrails, balustrades and supporting structures. Consistent dimensions improve safety and comfort. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

13. Structural Systems

Core Concepts

Structures transfer loads safely to the ground. Loads may include dead loads, imposed loads, wind and other environmental actions. Common structural systems include load-bearing masonry, reinforced concrete frames, steel frames, timber frames and composite systems.

Technical Application and Study Notes

Reinforced concrete combines concrete's compressive capacity with steel reinforcement's tensile capacity. Proper reinforcement, concrete cover, compaction and curing are essential. Architectural and structural drawings must be coordinated to avoid conflicts.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

13. Structural Systems — Extended Notes

Detailed Discussion

Structures transfer loads safely to the ground. Loads may include dead loads, imposed loads, wind and other environmental actions. Common structural systems include load-bearing masonry, reinforced concrete frames, steel frames, timber frames and composite systems. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

14. Building Services

Core Concepts

Building services make buildings usable, safe and comfortable. They include water supply, drainage, electrical systems, lighting, ventilation, air conditioning, fire protection and communications.

Technical Application and Study Notes

Services need adequate space for pipes, ducts, cables, equipment and maintenance. Early coordination between architectural and engineering drawings reduces clashes. Service systems should also allow inspection and repair.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

14. Building Services — Extended Notes

Detailed Discussion

Building services make buildings usable, safe and comfortable. They include water supply, drainage, electrical systems, lighting, ventilation, air conditioning, fire protection and communications. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

15. Plumbing and Sanitary Installation

Core Concepts

Water supply systems deliver water to fixtures and equipment through pipes, valves, tanks and pumps where required. Drainage removes wastewater and should be designed to prevent contamination and unpleasant odours.

Technical Application and Study Notes

Sanitary fixtures include water closets, wash basins, sinks, showers and floor drains. Rainwater is collected through gutters and downpipes and should be discharged safely to prevent erosion, flooding and dampness.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

15. Plumbing and Sanitary Installation — Extended Notes

Detailed Discussion

Water supply systems deliver water to fixtures and equipment through pipes, valves, tanks and pumps where required. Drainage removes wastewater and should be designed to prevent contamination and unpleasant odours. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

16. Electrical Installation and Lighting

Core Concepts

Electrical systems distribute power to lighting, sockets, equipment and specialized installations. Safety depends on appropriate protection, isolation, earthing and correctly selected conductors.

Technical Application and Study Notes

Lighting may be natural or artificial. Good daylighting considers orientation, window size, shading and glare. Efficient lighting, controls and appropriate equipment can reduce energy consumption.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

16. Electrical Installation and Lighting — Extended Notes

Detailed Discussion

Electrical systems distribute power to lighting, sockets, equipment and specialized installations. Safety depends on appropriate protection, isolation, earthing and correctly selected conductors. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

17. Ventilation, Air Conditioning and Acoustics

Core Concepts

Ventilation supplies fresh air and removes heat, moisture, odours and pollutants. It can be natural, mechanical or mixed. Air conditioning can control temperature and humidity depending on the system.

Technical Application and Study Notes

Acoustics deals with sound transmission, absorption, reflection and reverberation. Space planning, insulation, absorptive materials and appropriate construction can improve speech intelligibility and reduce unwanted noise.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

17. Ventilation, Air Conditioning and Acoustics — Extended Notes

Detailed Discussion

Ventilation supplies fresh air and removes heat, moisture, odours and pollutants. It can be natural, mechanical or mixed. Air conditioning can control temperature and humidity depending on the system. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

18. Environmental Design and Sustainability

Core Concepts

Sustainable architectural technology aims to reduce resource use and environmental impact while maintaining health and comfort. Strategies include orientation, shading, natural ventilation, daylighting, insulation, efficient systems and durable materials.

Technical Application and Study Notes

Water conservation can use efficient fixtures, rainwater management and suitable landscaping. Material selection should consider durability, availability, maintenance and environmental effects. Passive design is especially important in warm climates.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

18. Environmental Design and Sustainability — Extended Notes

Detailed Discussion

Sustainable architectural technology aims to reduce resource use and environmental impact while maintaining health and comfort. Strategies include orientation, shading, natural ventilation, daylighting, insulation, efficient systems and durable materials. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

19. Site Planning and Landscaping

Core Concepts

Site planning organizes buildings, roads, parking, pedestrian routes, drainage, services and open spaces. Good planning responds to topography, climate, access and surrounding development.

Technical Application and Study Notes

Landscaping may include vegetation, paving, walls, outdoor lighting and drainage. Plant selection should suit local conditions and maintenance resources. Pedestrian and emergency routes should remain safe and accessible.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

19. Site Planning and Landscaping — Extended Notes

Detailed Discussion

Site planning organizes buildings, roads, parking, pedestrian routes, drainage, services and open spaces. Good planning responds to topography, climate, access and surrounding development. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

20. Building Regulations, Codes and Safety

Core Concepts

Building regulations establish minimum requirements for health, safety, structural adequacy, fire protection, accessibility and other aspects of construction. Designers must use the current requirements applicable to their project location.

Technical Application and Study Notes

Fire safety may include escape routes, compartmentation, alarms, extinguishers, fire-resistant construction and emergency access. Construction hazards include falls, excavation collapse, electrical risks, machinery, dust, noise and falling objects. Safety planning, training and protective equipment are essential.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

20. Building Regulations, Codes and Safety — Extended Notes

Detailed Discussion

Building regulations establish minimum requirements for health, safety, structural adequacy, fire protection, accessibility and other aspects of construction. Designers must use the current requirements applicable to their project location. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

21. Construction Methods and Site Management

Core Concepts

Construction normally progresses through site preparation, setting out, substructure, superstructure, roofing, services, finishes, external works, testing and handover. Actual sequencing depends on the project.

Technical Application and Study Notes

Site management coordinates labour, materials, plant, access, storage, safety, quality and progress. Quality control uses approved materials, inspection, testing and documentation. Preventing defects is generally cheaper than correcting them after completion.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

21. Construction Methods and Site Management — Extended Notes

Detailed Discussion

Construction normally progresses through site preparation, setting out, substructure, superstructure, roofing, services, finishes, external works, testing and handover. Actual sequencing depends on the project. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

22. CAD and Digital Architectural Technology

Core Concepts

Computer-Aided Design allows accurate drafting, editing, dimensioning, annotation and documentation. CAD improves consistency and makes revisions easier.

Technical Application and Study Notes

Digital workflows may progress from site information to concept design, coordinated drawings, details, schedules and construction documentation. Software is a tool; users still require knowledge of drawing standards, construction and design.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

22. CAD and Digital Architectural Technology — Extended Notes

Detailed Discussion

Computer-Aided Design allows accurate drafting, editing, dimensioning, annotation and documentation. CAD improves consistency and makes revisions easier. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

23. Building Information Modelling (BIM)

Core Concepts

BIM is a digital approach for creating and managing information about built assets. Models can contain geometry and information about materials, systems, quantities, performance and maintenance.

Technical Application and Study Notes

BIM supports coordination between architecture, structure and services and can help identify clashes before construction. Other uses include visualization, quantity information, planning and facility management.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

23. Building Information Modelling (BIM) — Extended Notes

Detailed Discussion

BIM is a digital approach for creating and managing information about built assets. Models can contain geometry and information about materials, systems, quantities, performance and maintenance. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

24. Modern Construction Technologies

Core Concepts

Prefabrication produces building components away from their final location and assembles them on site. Modular construction uses standardized units or panels. These methods can improve speed and quality when logistics are properly planned.

Technical Application and Study Notes

Digital fabrication, automation and construction robotics are developing technologies. Their suitability depends on equipment, regulations, materials, skills, transportation and project scale.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

24. Modern Construction Technologies — Extended Notes

Detailed Discussion

Prefabrication produces building components away from their final location and assembles them on site. Modular construction uses standardized units or panels. These methods can improve speed and quality when logistics are properly planned. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

25. Nigerian Building Construction Practice

Core Concepts

Construction practice in Nigeria is influenced by climate, materials, workmanship, infrastructure, urban growth and economic conditions. Architectural design should respond to local environmental and social requirements.

Technical Application and Study Notes

In warm climates, shading, cross ventilation, appropriate orientation, roof design and control of solar heat gain are important. Buildings must also manage heavy rainfall and drainage. Local materials can be useful when their performance is appropriate and quality is controlled.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

25. Nigerian Building Construction Practice — Extended Notes

Detailed Discussion

Construction practice in Nigeria is influenced by climate, materials, workmanship, infrastructure, urban growth and economic conditions. Architectural design should respond to local environmental and social requirements. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

26. Construction Management and Cost Awareness

Core Concepts

Construction management coordinates time, cost, quality, resources, procurement, labour and risk. Architectural decisions affect both construction cost and long-term operating cost.

Technical Application and Study Notes

Good documentation includes drawings, specifications, schedules, inspection records, progress reports and site instructions. Clear documentation improves communication, accountability and project control.

Key points: understand the definition, identify the main components, explain the purpose of each component, and relate the topic to practical building construction.

26. Construction Management and Cost Awareness — Extended Notes

Detailed Discussion

Construction management coordinates time, cost, quality, resources, procurement, labour and risk. Architectural decisions affect both construction cost and long-term operating cost. In practice, the topic must be considered together with site conditions, user requirements, construction methods, cost, safety and maintenance. Students should learn to explain not only what a building component is, but why it is used and how it affects other components.

Practical Example

For a small residential building, the designer can begin with site investigation and space planning, select an appropriate foundation based on ground conditions, coordinate wall and structural systems, provide suitable openings, design roof drainage, and coordinate water, drainage and electrical services. The final drawings should communicate dimensions, materials, construction details and relevant notes.

Revision Focus

Define the major terms in this chapter. List the main types or components. Explain their functions. State advantages and limitations where relevant. Describe at least one practical application in building construction.

27. Revision Questions and Answers

What is architectural technology?

Answer: It is the application of technical and construction knowledge to the design, detailing, construction and performance of buildings.

What is a foundation?

Answer: A building component that transfers structural loads safely to the ground.

What is the difference between a plan and an elevation?

Answer: A plan represents a horizontal arrangement, while an elevation shows a vertical face.

Mention five building materials.

Answer: Concrete, steel, timber, masonry and glass.

What is BIM?

Answer: Building Information Modelling, a digital information-based approach to built assets.

Why is site investigation important?

Answer: It provides information about ground, drainage, topography, access and other site conditions affecting design and construction.

Mention four building services.

Answer: Water supply, drainage, electrical installation and ventilation.

What is sustainable design?

Answer: Design that reduces resource consumption and environmental impacts while maintaining useful building performance.

Mention four common roof forms.

Answer: Gable, hip, flat and mono-pitch.

Why is construction safety important?

Answer: It protects workers and the public, reduces accidents and supports safe and efficient project delivery.

GLOSSARY

Architectural technology: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Beam: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Column: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Foundation: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Elevation: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Floor plan: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Section: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Lintel: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Damp-proof course: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

BIM: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

CAD: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Rafter: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Ventilation: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Thermal insulation: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Sustainability: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Load-bearing wall: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Site investigation: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Prefabrication: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Modular construction: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

Building services: A key architectural/construction term that students should be able to define and apply correctly in technical discussions and drawings.

REFERENCES AND FURTHER READING

Ching, F. D. K. Building Construction Illustrated. Wiley.

Ching, F. D. K. Architecture: Form, Space, and Order. Wiley.

Allen, E., & Iano, J. Fundamentals of Building Construction: Materials and Methods. Wiley.

Eastman, C., Teicholz, P., Sacks, R., & Liston, K. BIM Handbook. Wiley.

International Code Council. International Building Code.

Neufert, E., & Neufert, P. Architects' Data. Wiley-Blackwell.

United Nations Environment Programme. Resources on buildings, construction and sustainable development.

Current Nigerian building regulations, standards and applicable local authority requirements should be consulted for project work.